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Perceptual Grouping in Contrastive Vision-Language Models

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Abstract

Recent advances in zero-shot image recognition suggest that vision-language models learn generic visual representations with a high degree of semantic information that may be arbitrarily probed with natural language phrases. Understanding an image, however, is not just about understanding what content resides within an image, but importantly, where that content resides. In this work we examine how well vision-language models are able to understand where objects reside within an image and group together visually related parts of the imagery. We demonstrate how contemporary vision and language representation learning models based on contrastive losses and large web-based data capture limited object localization information. We propose a minimal set of modifications that results in models that uniquely learn both semantic and spatial information. We measure this performance in terms of zero-shot image recognition, unsupervised bottom-up and top-down semantic segmentations, as well as robustness analyses. We find that the resulting model achieves state-of-the-art results in terms of unsupervised segmentation, and demonstrate that the learned representations are uniquely robust to spurious correlations in datasets designed to probe the causal behavior of vision models.

1. Introduction

Learning a representation for visual imagery requires resolving not only what resides within an image, but also where that information resides [72]. In many applications, knowledge of *where* information resides is sometimes more important than a precise description of the content [33, 98]. Hence, our ability to learn more generic and robust visual representations requires learning the geometry of visual semantics, and how visual information may be grounded by specific regions of the visual field.

While recent vision-language models trained under weak supervision demonstrate a remarkable ability to learn generic and transferable visual representations [50, 85, 117, 24], they showcase a profound inability to associate visual content with individual objects (Fig. 1, bottom row). In other words, models trained on large weakly-supervised data have a limited ability to group together visually related content [36]. Because the representations have a poor understanding of *where* an object resides, they easily conflate background with foreground content. Hence, the learned representations are unable to learn the spatial layout of a scene [97, 101], and are susceptible to learning spurious correlations between a semantic label and extraneous content [91, 65].

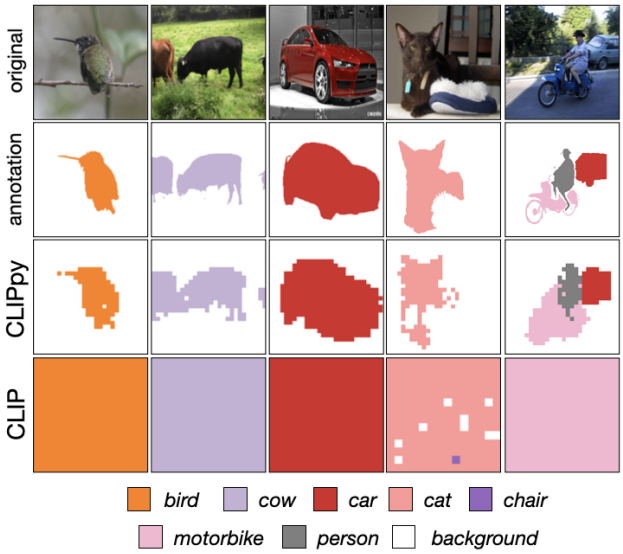


Figure 1: Semantic localization in contrastive VLMs. We measure the ability of vision-language models to predict a label at each spatial position in a zero shot manner based on the similarity of location tokens to the corresponding language tokens on selected examples. CLIP / ALIGN [50, 85] have minimal understanding of the spatial location of individual objects (row 4). Our proposed CLIPpy (row 3) predicts the label at locations that correspond closely to human annotation for semantic segmentation (row 2). All predictions were performed with no access to any segmentation data during training or inference. More visualizations in App. B.

Recent work [113, 114] attempts to bridge this gap through grouping mechanisms under the same weakly supervised training paradigm, but focus more on foreground

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objects (neglecting background classes). Another direction is task specific unsupervised fine-tuning [126, 26] which loses the generic and transferable nature of these representations.

In this work, we explore vision-language models that learn from similar weakly labeled data, but a) retain the generic and transferable nature of features, and b) learns where all (background and foreground) visual content resides within an image. Unlike previous attempts using grouping specific architectures [113, 114] or dense human annotations [36, 38, 57], we explore a minimal set of modifications to existing CLIP models [85] that leads to grouping of visual imagery while retaining their weakly supervised and scalable training procedure. We find that two small adjustments – employing specific pretraining strategies and adjusting spatial feature aggregation – results in models that are equally effective in zero-shot image recognition, but also retain spatial information regarding object locations (see Fig. 1, 3rd row).

The resulting model termed CLIPpy exhibits *perceptual grouping* – that is, the ability to select and combine related visual signals into semantically meaningful regions [110, 72, 89]. Endowing models with perceptual grouping – whether in a bottom up (based solely on visual content) or top down (guided by external information, language in this case) manner – in learned representations has been a long standing goal in computer vision [70, 71]. In this work, our key contributions are as follows:

- Identify systematic failure of contrastive vision-language models [85, 50] to properly identify where objects reside within an image, and group semantically related content.
- Design a minimal set of changes to endow these model with perceptual grouping, resulting in state-of-the-art zero-shot segmentation *without* training on *any* segmentation data or performing task specific fine-tuning.
- Emergence of localization ability in our models uniquely leads to robustness to counterfactual manipulations. The degree of robustness matches if not surpasses previous state-of-the-art supervised learning methods employing specialized training methodologies.

2. Related Work

Vision-language models for grounding. Contrastive language image pre-training [85] (CLIP) led to a range of follow up work performing open-vocabulary detection [38, 51, 58, 59, 120, 28] or segmentation [36, 57, 122]. While these methods leverage dense human annotations for training, an alternate line of works [113, 114, 126, 115, 22] attempt to learn alignment between regions of images and language with only image level noisy captions for supervision. Their weak supervision allows better scalability (to more data) leading to learning more generic and transferable representations. In fact, multiple such works [113, 114, 126, 26, 57] perform zero-shot semantic segmentation. However, unlike [113, 114] geared to segment a fixed count of foreground objects, our proposed CLIPpy can better segment arbitrary object counts and background classes. In contrast to [126] using generic image level features, CLIPpy explicitly learns local features during training. Moreover, CLIPpy requires no dense human annotations or task-specific fine-tuning in contrast to [26, 57]. We also highlight how [113, 114, 26] perform grouping independent of language at inference - however CLIPpy can group conditioned on language, capturing variable object boundaries for different language prompts.

Component	CLIP [85]	CLIP†	CLIPpy
Image Backbone	ViT-B/16	ViT-B/16	ViT-B/16
Text Backbone	T-B	T-5	T-5
Image Init	Random	Random	DINO
Text Init	Random	Random	Sent T-5
Image Pooling	CLS	CLS	Max
Text Pooling	Avg	Avg	Avg
Dataset	300M*	CC-12M	CC-12M
VOC mIoU (%)	16.4	17.5	**50.8** (+33.3)
VOC JS (%)	28.6	37.3	**47.5** (+10.2)

Table 1: We highlight the minimal differences of CLIPy from CLIP. CLIP† is our implementation following train settings identical to CLIPpy. *indicates OpenAI private data.

Multiple **contemporary works** also explore similar directions as CLIPpy, leveraging pre-trained vision-language models for various grouping tasks under weak supervision (no pixel level annotation) [123, 68, 13, 75, 9, 52]. Combining self-supervised methods that emerge grouping [12] with CLIP models [85] for cross-modal alignment is explored in [123] gaining notable improvements at object boundaries. A clustering mechanism containing learnable centres similar to [113] is combined with reconstruction and super-pixel alignment losses to achieve grouping in [68]. Learning decoder networks over a frozen CLIP backbone [85] with text to image patch similarity losses are explored in [13, 75] resulting in similar grouping behaviour. In contrast to these methods utilizing contrastive vision language training to emerge grouping, recent works [9, 52] also showcase how text-to-image generative models (particularly Stable Diffusion [90]) can be leveraged to perform visual grouping.

Zero-shot semantic segmentation. A form of top-down grouping, this relatively new task [124, 48, 111, 8, 79, 45, 60, 3, 95] attempts to segment unseen classes, usually after a supervised training phase often involving dense annotation based supervision. Following two early representative works [111, 8], most later approaches [60, 39, 40, 53, 95, 102] formulate the task as a pixel-level zero-shot classification problem with a closed set vocabulary. While CLIPpy follows a similar pixel based formulation, in contrast, our method requires no dense human annotations for supervision, no task specific fine-tuning, and is open-vocabulary. Recent work [26, 58] also explores region-level classification leveraging pre-trained CLIP models [85], but unlike CLIPpy perform grouping independent of language during inference.

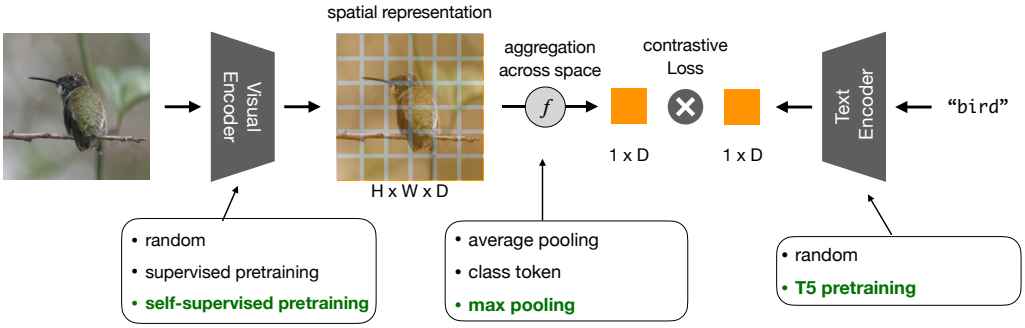


Figure 2: Architecture diagram. Images and captions are separately embedded into Euclidean spaces, where image features are spatially aggregated. A contrastive loss trains the aggregated image embedding to be close to the caption embedding. We demonstrate that two minimal design decisions (indicated in green) are of paramount importance for CLIP [85] models to perform perceptual grouping under image-level weak supervision.

Unsupervised segmentation. Analogous to bottom-up grouping, these works perform class-agnostic segmentation within the visual modality with no explicit language alignment [12, 41, 31, 49, 73]. This topic has a long, rich history in human visual perception [110] and computer vision [70], and has been explored as means of generalizing to new visual domains [84, 71]. It is this goal that most closely inspires our work. Early efforts group pixels based on known spatially-local affinities [19, 96, 88], with subsequent methods leading to region proposal networks for object detection [103] and advances in semantic segmentation [1]. Recent methods employ self-supervision to learn perceptual grouping [18, 41] or object-centric groupings [29, 66, 109, 4, 44]. Our proposed CLIPpy demonstrates competitive performance, but additionally aligns groups to the language modality explicitly.

Learning robust visual representations. For a long time, ImageNet [23] accuracy was believed to provide a reasonable proxy for quality of learned visual representations [37, 55]. However, recent work highlights notable deficiencies in such learned representations [34, 87, 54] including sensitivity to low level textures, failure for domain shifts, and reliance on spurious correlations. These failures inspired a large literature to mitigate learning spurious correlations [91, 65, 2] by focusing on new optimization techniques. Progress on this issue may address parallel issues in fairness [21]. Resulting methods have largely focused on synthetic data, re-balancing data, and shaping learned embeddings [76, 65]. Nonetheless, theoretical results suggest pessimistic bounds unless additional structure informs the problem (see refs. in [91]). Therein, the structured output predictions of proposed CLIPpy provide another promising solution.

3. Methodology

We first set the stage by discussing established core architectures and the contrastive learning formulation. Next, we discuss modifications that are the focus of the analysis in this work. In particular, we discuss aggregation options, pre-training alternatives, and token sub-sampling.

3.1. Architecture and Training

We provide a quick overview of our architecture (Fig. 2). Consider a batch size N, spatial height H, spatial width W, and depth D. X is a tensor that has a shape of $[N, H, W, D]$ and is the output of an image encoder. Y is a tensor that is of shape $[N, D]$ and is the output of a text encoder.

Language Model. We employ a strong language model baseline derived from the transformer architecture [104] and implemented in T5 [86]. T5 models use an encoder-decoder architecture that is trained using a generative span corruption task, and have achieved state-of-the-art on a broad range of NLP tasks including GLUE [106] and Super-Glue [105]. We use the encoder only and discard the decoder part. We employ the T5-base which consists of 12 transformer layers, 12 attention heads, and 768 token channel dimensions.

Image Model. We explore two architectures for image featurization, CNN-based and Vision-Transformers, although we focus the majority of work on the latter. First, we employ the EfficientNet architecture [100] as a high performant CNN architecture, which has been used previously in vision-language models. The specifics of the meta-architecture were derived from considerations based on neural architecture search. Second, we employ the Vision Transformer (ViT) architecture [27]. We refer the reader to [27, 104] for details. Briefly, ViT is largely inherited from the NLP literature and consists of a hierarchical associative memory. Each layer, termed a transformer, is composed of a Multi-headed Self-Attention (MSA) layer followed by a 2-layer feed-forward multi-layer perceptron (MLP). The primary parameter of ViT is the patch size P specifying the $P \times P$ patch of pixels constituting a token in the architecture.

Contrastive Representation Learning. Let x_i and y_i denote the image and text embeddings (post aggregation) of the i'th example in the batch. A contrastive loss may be specified as the cross entropy across a batch [85, 50]. The cross entropy is calculated between a one-hot encoding specifying the correspondence between the image and text examples,

and a softmax-normalized distribution specifying the dot-product similarity between image and text embeddings.

$$L = \underbrace{-\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp(x_i^\top y_i/\tau)}{\sum_{j=1}^{N}\exp(x_i^\top y_j/\tau)}}_{\text{image-to-text}} + \underbrace{-\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp(y_i^\top x_i/\tau)}{\sum_{j=1}^{N}\exp(y_i^\top x_j/\tau)}}_{\text{text-to-image}}$$

The normalization for the image-to-text and text-to-image similarity is computed by summing over the potential matches (indexed by j) to the text and image examples within a batch, respectively. Note that τ is the temperature of the softmax for the normalization.

3.2. Aggregation

The goal of the aggregation method is to collapse the image embedding from a $[H, W, D]$ tensor to a D dimensional vector. **Average pooling** across space is an established technique for ensuring that the final embedding is independent of the image resolution [99, 67], and has been adopted for CNN-based architectures in vision-language models [50]. Alternatively, **maximum pooling** has been explored, in particular with success for point clouds [83] and image-audio [42]. Another approach typical for ViT borrowed from language modeling [25] is the **class token (CLS)**, which is prepended to the image patch tokens [27]. A class token learns an embedding that aggregates information across all patch tokens in order to predict the image label. The class token may be used to summarize the content for an entire image for ViT-based models [85, 12]. Subsequent work in vision-language models has explored learning pooling strategies [15, 115], heuristically selecting a set of similar neighbors [118] or learning attention-based mechanisms [117].

In this work we systematically explore these aggregation strategies. In early experiments we found that many complex strategies for aggregation yielded poor results (App. A.2). We found that the application of max pooling across the spatial dimensions – while extremely simple – was also by far the most effective (Sec. 4.5). We hypothesize that the success of max pooling may be due to the gradient updates being focused solely on a single spatial location, and not spread across all spatial dimensions.

Why Max Pooling? In particular, the max pooling operation allows pre-aggregation features (shaped $[N, H, W, D]$) to determine the spatial location for gradient updates at each step, conditioned on input images. Across different images containing a common object at different spatial locations, the model has to select a conservative and minimal set of spatial locations for gradient updates. At the same time, given the cross-modal contrastive train objective, the aggregated feature of each such image must be aligned towards a common language concept (i.e. related to the common object). We hypothesize that gradient updates at the common object's spatial location is the simplest optimization for the train objective in this case, leading to observed perceptual grouping.

3.3. Pretraining

Language Model. For better sentence level representation, we utilize pre-training from Sentence-T5 [78] which adapts a T5 encoder to sentence level embedding using a contrastive objective. We select Sentence-T5 over auto-regressive models such as [25, 7] because this contrastive loss is aligned to our setup. The model is trained on Stanford Natural Language Inference (SNLI) dataset with 275K examples focused on entailment questions [6, 32].

Image Model. We investigate initializing the image model with several methods. First, we investigate initializing the image model using **supervised pre-training** and removing the final layer for logistic regression [37, 55]. We next investigate **self-supervised methods** derived from self-distillation (e.g. [12]). We focused on this latter direction because such models demonstrated impressive performance in terms of localization [12, 41]. All image pre-training is performed on ImageNet-1K [23] dataset.

Suitable Visual Pre-training. The visual encoder representation space can be viewed as containing per-image features (post-aggregation) vs per-spatial location features (pre-aggregation). We hypothesize that semantics tied boundaries of this representation space should operate at the latter granularity to induce perceptual grouping. Furthermore, we suggest that initializations facilitating the former will detriment grouping behaviour. In particular, visual pre-training strategies separating image-level representations by semantics (e.g. supervised ImageNet pre-training) will diminish perceptual grouping. Self-supervised pre-training strategies focused on more granular within image representations (e.g. [12]) will tend to enhance perceptual grouping. This hypothesis is empirically validated in ablations (see Table 8).

3.4. Visual Token Sub-Sampling

Motivated by vision transformers' ability to process sequences of length different to train time, we generate higher resolution segmentations during inference by sampling more image patches. In order to increase robustness to such varying resolution, we utilize up to $2\times$ higher resolution images during training but randomly drop 80% of visual tokens to minimize additional compute overhead (similar to [43, 62]). While improving segmentations, this also provides training stability possibly due to its regularizing effect (see App. D).

3.5. Inference

CLIPpy performs inference under 3 different settings: a) classification, b) bottom-up grouping, and c) top-down grouping. On the visual modality, the first utilizes a spatially aggregated single per-image token while the latter two utilize sets of per-region tokens. Classification follows zero-shot analyses from [85] where the model is prompted at inference for a selection of labels (App. I for prompts). Bottom-up

grouping follows a form of spectral clustering inspired by [12] (refer to their demo). PCA on image features (from visual encoder pre-aggregation) gives top n(=8) principal components, which are used as cluster centers. Each of those same image features are assigned to one of the n clusters based on proximity (cosine similarity) to the centers, resulting in n clusters (or groups). Top-down grouping employs zero shot analysis similar to [85], but at each spatial location, using the per-region tokens. This is similar to [35] and generates predictions across space exploiting the transitive property of our aggregation operations.

4. Experiments

Experimental Setup. We train our models on two datasets: Conceptual Captions 12M (CC-12M) [14] and High Quality Image Text Pairs (HQITP-134M) consisting of 12 million and 134 million image-text pairs, respectively (App. C for details). For both datasets, text is tokenized, and images resized and center cropped to 224×224 pixels. We report results on EfficientNet-B5 employed by ALIGN [50], and ViT-B/16 employed by CLIP [85] although we focus more on the latter. We train models on 32 GPUs across 4 machines with PyTorch [80]. See App. D for more details. We evaluate across image classification, localization, and robustness tasks. For image classification, we employ the validation splits of ImageNet [23] and ImageNet-v2 [87], and for robustness we employ the test split of Waterbirds [91]. These datasets contain 1000, 1000, and 3 classes respectively. For segmentation tasks, we employ the validation splits of PASCAL VOC [30], ADE20K [125, 17], COCO [64], COCO (Obj) [64], and Cityscapes [20]. Each of these datasets contain 20, 150, 133, 80, and 27 labels, respectively.

Baselines for comparison. Given that most competitive baselines are trained on private datasets, we first attempt to reproduce results by training models on a corpus of image-text pairs. In more detail, we train on the public CC-12M dataset [14] to provide reproducible numbers and observe competitive performance given our data limitations. We also train on the larger HQITP-134M dataset to verify scalability.

We first measure the performance of CLIP [85] and ALIGN [50] on zero-shot image classification on ImageNet and ImageNet-v2. Table 2 highlights these results. We take this as a starting point for subsequent work. In the following experiments we attempt to address the following questions:

- What are the limitations of current vision-language models? (Fig. 1)
- Do we observe perceptual grouping in vision language models? (Tabs. 3, 4 and 6).
- How resilient are vision-language models to counterfactual manipulations? (Fig. 4).
- How important are each of the proposed model modifications? (Tabs. 7 to 10).

	Dataset	IN	IN-v2
ALIGN [50]	ALIGN-1800M	76.4	70.1
CLIP [85]	CLIP-400M	65.5	60.8
CLIP †	CC-12M	46.0	40.3
GroupViT [113]	CC-12M+YFCC	42.9	-
GroupViT †	CC-12M	25.6	23.8
CLIPpy	CC-12M	45.3	40.0
ALIGN †	HQITP-134M	51.1	45.6
CLIP †	HQITP-134M	61.4	56.4
CLIPpy	HQITP-134M	60.3	54.8

Table 2: CLIPpy achieves competitive zero-shot image recognition. IN and IN-v2 denote ImageNet and ImageNet-v2 accuracy, respectively. † indicates our implementation. [50] evaluated at 640×640; others evaluated at 224×224. CLIPpy shows ±0.5 and ±0.9 IN acc. (5 runs) on CC-12M and HQITP-134M, respectively.

4.1. Limitations of vision-language models

Visual representations learned in vision-language models exhibit an impressive ability to generalize across tasks [85, 50]. However they also exhibit a profound shortcoming – learned visual representations maintain minimal information about *where* an object resides, failing to properly recognize what parts of an image constitute an object.

Fig. 1 (bottom row) showcases failure of a CLIP model; namely, the model improperly conflates visual content not associated with an object with the actual object. This can be observed by measuring the similarity of each embedding at each spatial location with a label set using the method in [35] (Sec. 3.5). One consistently observes that the central object of interest is incorrectly predicted to reside at every spatial location. For instance, in the left example, the CLIP model predicts that a `bird` resides at every spatial location. In a CNN architecture, where spatial information is inherently preserved, we observe some improvement, but the larger issue of poor localization remains (see App. E for details).

This failure of vision-language models to properly understand the spatial organization of information is consistent with earlier observations. Ablation experiments in ViT models demonstrated that removing positional embeddings minimally detriments predictive performance [27, 77, 121, 97]. Without positional information, ViT models effectively learn representations as a "bag of image patches", ignoring the spatial organization.

In contrast, if we perform the same analysis on CLIPpy, we see that the model retains significant information about spatial information (Fig. 1, 3rd row). We take these visualizations as an impetus for further investigation. In particular, we start by quantifying the ability of the model to arbitrarily group together semantically related pixels, and compare this to previous works.



Figure 3: Visualizations of bottom-up grouping by CLIPpy. Each color represents one grouping learned on a given image.

4.2. Emergence of Bottom-Up Perceptual Grouping

Unsupervised segmentation performance is a direct measure of bottom up perceptual grouping. We apply CLIPpy at test time to perform semantic segmentation without prompting it for any labels [1]. Fig. 3 shows how the model visually groups semantically related regions of an image (see also Fig. 5 in App.) as the image embeddings naturally group into spatially distinct clusters mirroring the image structure. We emphasize that this analysis does *not* rely on text prompts *nor* segmentation labels, but merely emerges from the image features alone. Hence the model has learned to *group* perceptually related pixels merely based on the pixel content and associated image-level captions during training.

We quantify the accuracy of this bottom-up segmentation to capture known segmentations within annotated images. Following evaluation protocol in [12, 113], we compute the Jaccard Similarity (JS). JS here measures the average intersection over the union across all segmentation instances regardless of object category. Our results in Tabs. 3 to 5 demonstrate competitive performance by CLIPpy. In VOC, CLIPpy achieves 54.6% outperforming all previous models; in comparison, CLIP achieves 38.9%. Additionally, on two more challenging datasets we note how the model drops in performance relatively, perhaps indicative of more visually cluttered scenes (Tab. 4). Our intuition for CLIPpy improving over CLIP is that CLS and average pooling breaks spatial structure of features, mixing image-level features across features at all spatial locations. We take these results to indicate that CLIPpy perceptually groups semantically related content better than previous work, providing state-of-the-art results in unsupervised segmentation.

4.3. Top-down Grouping

We demonstrated that CLIPpy is able to perceptually group visual content within an image. Next, we ask how well this grouping corresponds to semantically meaningful labels. To measure the emergence of top-down grouping, we ask how well the perceptual grouping of the model may be steered by embeddings from the language model. We test this by comparing the model's ability to perform zero-shot semantic segmentation across four datasets. Note that all of our results and comparisons are solely restricted to models trained on *no* segmentation annotations[2].

Fig. 1 provides a visualization of the predicted zero-shot segmentations (see also App. B), and Tab. 6 quantifies the results using mean intersection over union (mIoU). CLIPpy outperforms all other approaches on semantic segmentation when trained on the same datasets, both for CC-12M and HQITP-134M. We view our datasets in two categories: ADE20K and COCO contain numerous background classes while VOC and COCO (obj) contain only foreground object classes. We particularly highlight the notable performance improvement of CLIPpy for the former datasets. Moreover, in comparison to CLIP and ALIGN baselines, CLIPpy achieves significant improvements. We also replicate these baselines on the largest possible dataset within our compute budget (HQITP), for comparison on a common dataset.

[1] We perform PCA clustering (see Sec. 3.5). GroupViT [113] & DINO [12] employ 8 & 6 feature vectors based on their model architectures. Our visualizations employ 8 feature vectors (cluster centers).

[2] In App. F, we provide a summary of other zero-shot semantic segmentation results. Some of these prior results achieve superior performance, but we note that all of these methods were trained explicitly on various forms of segmentation masks, if not segmentation labels, often with task specific fine-tuning in contrast to the generic & unsupervised nature of CLIPpy.

	Dataset	Train	SSP	VOC
DeiT [12]	ImageNet	class	✗	24.6
MoCo [113]		self	✓	28.2
DINO [12]		self	✓	45.9
DSM [73]		self	✓	37.2
COMUS [119]		self	✓	47.3
DINO [113]	CC-12M & YFCC-100M	self	✓	41.8
CLIP [113]		text	✗	28.6
GroupViT [113]		text	✗	51.8
CLIP †	CC-12M	text	✗	37.3
GroupViT †		text	✓	42.8 (+5.5)
CLIPpy		text	✓	**47.5** (+10.2)
CLIP †	HQITP-134M	text	✗	38.9
CLIPpy		text	✓	**54.6** (+15.7)

Table 3: CLIPpy effectively performs bottom-up grouping. We report Jaccard Similarity, an instance average of IoU between predicted and annotated segmentations, independent of object labels. †denotes our implementations. SSP indicates the use of self-supervised visual pre-training.

	Dataset	ADE20K	COCO
CLIP †	CC-12M	22.9	20.4
CLIPpy		**28.9** (+6.0)	**26.0** (+5.6)
CLIP †	HQITP-134M	24.2	21.6
CLIPpy		**29.5** (+5.3)	**27.2** (+5.6)

Table 4: More bottom-up grouping: CLIPpy improves Jaccard Similarity across datasets.

Method	JS
IIC [49]	6.4
MDC [11]	7.1
PiCIE [18]	12.3
STEGO [41]	21.0
CLIPpy (ours)	**22.3**

Table 5: More bottom-up grouping: CLIPpy achieves competitive Jaccard Similarity (JS) on the Cityscapes Dataset 27 class segmentation setup [41].

	Arch	Dataset	SSP	ADE20K	COCO	VOC	COCO (obj)
GroupViT [113]	ViT		✗	-	-	41.1	-
GroupViT †	ViT		✓	6.2	12.7	40.1	17.5
MaskCLIP † [126]	ViT	CC-12M	✗	6.8	8.1	22.1	13.8
OVS [114]	ViT		✓	7.1	-	44.6	25.1
CLIP †	ViT		✗	5.0	7.8	17.5	13.2
CLIPpy	ViT		✓	**13.1** (+8.1)	**23.8** (+16.0)	**50.8** (+33.3)	**28.5** (+15.3)
ALIGN [35]	CNN	ALIGN-1800M	✗	9.7	15.6	-	-
CLIP [85]	ViT	CLIP-400M	✗	5.8	8.7	16.4	14.5
ALIGN †	CNN		✗	7.5	14.4	29.7	-
CLIP †	ViT	HQITP-134M	✗	5.1	8.0	18.1	14.5
CLIPpy	ViT		✓	**13.5** (+8.4)	**25.5** (+17.5)	**52.2** (+34.1)	**32.0** (+17.5)

Table 6: CLIPpy provides competitive localization with no segmentation or location annotations. All models trained without any segmentation annotations. Results grouped by training dataset (bold highlights best per dataset). Numbers are mean IoU. † indicates our implementation. SSP indicates image self-supervised pre-training to visual encoder.

These results on HQITP also indicate clear performance improvements from CLIPpy.

GroupViT [113] and OVS [114] provide important points of comparison. These models use custom ViT architectures specific to grouping, are trained on common datasets (containing image-text pairs), and are designed to perform perceptual grouping by optimizing discretized attention masks. We draw attention to the clear performance improvements of CLIPpy over these methods across all datasets. We also highlight that OVS [114] uses pre-training strategies similar to ours. Our implementation of GroupViT also utilizes similar pre-training following [114]. We take these results to mean that our simple changes to existing vision-language models uncover powerful localization information[3].

4.4. Perceptual grouping may improve robustness

We have observed how parsimonious changes to vision-language models result in state-of-the-art unsupervised and zero-shot semantic segmentation. In this section, we ask how the resulting perceptual grouping may be exploited to improve the robustness of image understanding. A large literature has consistently observed that models systematically underperform under domain shifts [87]. For instance, CLIP, ALIGN, and CLIPpy underperform on ImageNet-v2 versus ImageNet (Tab. 2). Another means of assessing robustness is to measure how well a model *causally* predicts the label from the appropriate input variates [81, 82]. To probe for causal dependencies, one can measure model performance to counterfactual examples where an input is selectively manipulated in order to test for sensitivity to spurious correlations.

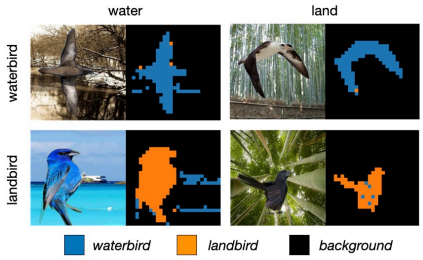
A common formulation for this problem is to artificially synthesize a malicious dataset where a trained model may correlate inappropriate image features to predict a label [112, 74, 46, 2]. A large class of supervised learning algorithms have been developed to train on these datasets[4] with the aim of mitigating such spurious correlations [91, 65, 76]. One common synthetic benchmark is *Waterbirds* [91] which places segmentations of birds in front of a background of land or water. The goal of any prediction system is a two-way classification of whether or not a bird is from the `waterbird` or `landbird` category. What makes this problem particularly challenging is when the background is not commensurate with the type of bird. For instance, a trained model may be prone to predict the type of bird due to the presence of water in the background in lieu of the visual appearance of the actual bird.

We first asked how our baseline CLIP model performs on this task when presented with a zero-shot three-way classification task (App. H for inference procedure). Model performance depends heavily on the background (Fig. 4 centre). For instance, the prediction accuracy of `waterbirds` drops by Δ=32.1% (80.2 $\rightarrow$ 48.1) in the presence of an incommensurate background. Clearly, the baseline CLIP model performs zero-shot prediction by relying on features from the background. We note that open-source CLIP [85] has similar trends (see App. Tab. 11).

We next asked how CLIPpy performs given that it exhibits a unique ability to discriminate the spatial locations of objects. Fig. 4 shows selected examples from each class colored by the prediction at each spatial location. Clearly, the model is able to discriminate which locations correspond to each category. We quantify model accuracy across each task, and find the model far less sensitive to the background. For instance, in the case of waterbirds, CLIPpy accuracy, while slightly less than the baseline CLIP model, only drops by Δ

[3]We note that even removing all pretraining and solely training on CC-12M still retains notable grouping performance (Tab. 8).

[4]Synthetic datasets are deliberately constructed to contain a class imbalance such that a minority class may be particularly prone to systematic worse performance. Consequently, experimenters have focused on the worse-case performance on the minority class [91, 65]. Our work is instead focused on the domain gap to target the degree to which spurious correlations inappropriately influence predictions.



CLIP	water	land	Δ
waterbird	80.2	48.1	**-32.1**
landbird	38.8	71.7	**-32.9**

CLIPpy	water	land	Δ
waterbird	76.9	74.9	**-2.0**
landbird	80.0	84.1	**-4.1**

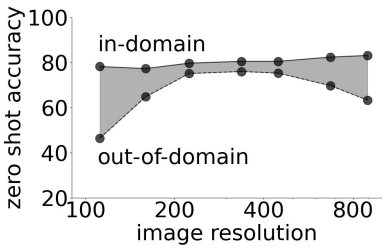


Figure 4: Perceptual grouping mitigates sensitivity to spurious correlations. (left) Selected segmentation examples by CLIPpy of waterbirds and landbirds on each background. (centre) Accuracy on the *test* split (5794 examples) of Waterbirds on CLIP and CLIPpy evaluated at 448×448 resolution. The domain gap Δ reports the drop in accuracy between on and off diagonal entries within a row. (right) Zero shot accuracy of CLIPpy across image resolution for landbirds on land (top) and water (bottom). Note log axis. Shading highlights Δ.

= 2.0% (76.9 → 74.9) in spite of the background change (Fig. 4 right). Interestingly, the domain gap Δ is minimal (∼4%) around a broad range of image input resolutions centered about the training resolution of the model (Fig. 4). Hence, CLIPpy, while still susceptible to some spurious correlations, is far more robust than a standard vision-language model.

As points of comparison, all prior work train a supervised model on the training split. In contrast, our predictions are zero-shot, and we do not use the training set. This difference makes a direct comparison of the raw accuracy difficult. That said, the best supervised training methods achieve a domain gap Δ of 4% to 8% (Tab. 1 and priv. correspondence, [65]), comparable to our results. We take these results to indicate that our zero-shot approach leveraging perceptual grouping provides another approach for addressing spurious correlations and learning robust image features.

4.5. Ablation Studies

We next perform experiments to demonstrate how individual factors in CLIPpy led to improved localization.

We first explore the effect of pre-trained representations. In Tab. 7, we freeze each of the backbones with self-supervised pre-training [12] for the image backbone and sentence T5 pre-training [78] for the text backbone. Our ablations indicate that the pre-trained weights alone do not contribute to the strong perceptual grouping of CLIPpy: our modified training process is necessary. In fact, both classification and semantic segmentation performance is affected negatively by freezing either backbone.

We also explore how alternate or no pre-training effects overall performance. Table 8 explores the selective removal of pre-training on the image model, language model or both. All models employ maximum pooling aggregation across spatial locations. Again, we see that CLIPpy exhibits significant drops in both zero-shot image recognition and localization by selectively dropping out each pre-training step. For instance, model performance drops from 42.3% to 25.6% top-1 accuracy. Likewise, the semantic segmentation mIoU drops from 50.8% to 23.5% accuracy. As expected, ImageNet supervised pre-training improves ImageNet top-1 accuracy, but interestingly leads to significant drops in grouping performance. For bottom-up segmentation, initializing from pretrained models benefits from scaling up the joint training data (Tab. 8 vs. 13). We suspect that these results indicate how each initialization provides valuable prior information not readily available in joint training for eliciting strong grouping properties, while also demonstrating the need for our training mechanism to emerge such grouping behaviour.

We next ablate the choice of aggregation mechanism. CLIPpy employs a maximum operation over all spatial locations. We likewise train models performing spatial averaging or employing a class token. We present these results in

Aggregation	I-F	T-F	IN (Acc)	VOC (mIoU)
Cls	✓	✗	39.9	3.4
Max	✓	✗	24.2	10.4
Max	✗	✓	35.9	29.5
Max	✗	✗	**42.3**	**50.8**

Table 7: Ablation on freezing pre-trained backbones: We report Top-1 accuracy (%) for ImageNet (IN) and mean IoU for VOC. I-P stands for image backbone pooling, I-F stands for image backbone frozen, and T-F stands for text backbone frozen.

Dataset	Image Init	T5 Init?	ImageNet Accuracy	Pascal VOC mIoU	Pascal VOC Jaccard
CC-12M	DINO	✓	42.3	**50.8**	**47.5**
	IN-1K	✓	53.3	22.5	43.3
	random	✓	28.9	32.9	43.6
	DINO		34.1	44.3	47.2
	IN-1K		44.5	20.0	42.2
	random		25.6	23.5	43.1

Table 8: Ablation on alternate pre-training: We report Top-1 accuracy (%) for ImageNet and mean IoU & Jaccard Similarity for VOC. Image encoder is initialized with DINO, supervised training on ImageNet-1K, or random weights. Text encoder is initialized with Sentence T5 or random weights. Parallel ablations using HQITP-134M in App. A.3.

Dataset	Aggreg.	ImageNet	Pascal VOC	
		Accuracy	mIoU	Jaccard
CC-12M	Max	42.3	**50.8**	**47.5**
	Avg	44.0	11.6	38.1
	Cls	46.0	4.0	40.4
HQITP-134M	Max	59.0	**50.1**	**54.6**
	Avg	60.0	17.9	40.5
	Cls	60.2	4.1	41.3

Table 9: Ablation across aggregation methods: We report Top-1 accuracy (%) for ImageNet and mean IoU & Jaccard Similarity for VOC. Global max pooling (Max), global average pooling (Avg), and class token (Cls) alternatives are explored. All models initialized with the same pre-trained features.

TSS	IN	VOC	COCO	ADE20K
✗	45.3	50.9	23.5	12.6
✓	**45.6**	**51.8**	**24.1**	**13.4**

Table 10: Ablation on token sub-sampling: We report top-1 accuracy (%) for ImageNet (IN) and mean IoU for the three segmentation datasets (VOC, COCO, ADE20K). TSS stands for token sub-sampling.

Tab. 9. The standard procedures of class token and average pooling result in similar performance on zero-shot classification on ImageNet, but notable reductions in mIoU on VOC semantic segmentation. For instance, in the model trained with CC-12M, mIoU on VOC drops from 50.8% to 4.0% representing a relative drop of 91.3%. Similarly, in the case of bottom-up grouping on the same dataset, we demonstrate a 10 point drop in JS. We use these results to highlight the significant role played by the aggregation mechanism in inducing observed grouping properties.

We finally explore the effect of proposed token sub-sampling in Tab. 10. Improvements in classification and semantic segmentation performance across datasets verify its role in boosting performance.

5. Discussion

In this work we demonstrated how contrastive vision-language models have a profound lack of understanding object location. We described a minimal set of changes to existing vision-language models by modifying the aggregation method, introducing optimal pre-training strategies, and train-time token sub-sampling techniques to endow the model with both bottom-up and top-down perceptual grouping. We emphasize that our changes are minimal but sufficient to match if not exceed the performance of custom-built architectures [113, 114] in achieving perceptual grouping. We demonstrate that our resulting model provides state-of-the-art results in terms of unsupervised segmentation, and competitive results in terms of zero-shot semantic segmentation – even though the model has been afforded *no* segmentation annotations whatsoever. Finally, we demonstrate the utility of these representations by demonstrating how perceptual grouping may be leveraged to learn visual features that are robust to spurious correlations.

We take these results to indicate that contrastive vision-language models may provide the emergence of perceptual grouping without supervision. We do see limitations in this approach as segmentation suffers with increasing visual clutter and label cardinality (e.g. ADE-20K). We suspect that recent advent of larger-scale open datasets [92, 10] and advances in self-supervised learning [41, 73] may offer opportunities to demonstrate further benefits for endowing models with perceptual grouping. We also note the possibility of biases in our training data that may be reflected in our models.

Reproducibility Statement

We built a codebase derived from OpenAI CLIP source code (`https://github.com/openai`). Source code changes to accommodate our modifications to architecture and training were minimal, and are all documented in Sec. 3. We employed pretrained *Sentence T5 Base* language models from HuggingFace (`https://huggingface.co`) and image models from [12, 27]. The CC-12M dataset was downloaded from [14] and provides a publicly reproducible benchmark. HQITP-134M can not be publicly released due to copyright issues. All reported numbers also contain an equivalent version for a model trained only on CC-12M to enable reproducibility.

Ethics Statement

We describe a minimal set of changes to vision-language models to endow these models with perceptual grouping and localization information. Our work contributes to a large literature for how to build more performant and generalizable vision models. We use both public and private computer vision datasets and leverage pretrained language and image models for our experiments. Although we believe our code and model architecture to contain no inherent bias, both the public and private data we employ may contain such biases. Any trained model should thus be approached and deployed with caution to ensure that that all fairness and bias issues are properly addressed.

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